

PET Quantification-Preserving Registration for Longitudinal Whole-Body PSMA PET/CT

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Abstract. We investigate how PET biomarker preservation and bone rigidity affect longitudinal whole-body PSMA PET/CT registration. The study uses a LapIRN pipeline with an affine stage and test-time instance optimization. The planned evaluation separates the contributions of PET preservation and rigidity, compares existing rigidity penalties, and characterizes the trade-off between anatomical alignment and lesion-level quantification preservation. [TODO: Replace this prospective abstract after the independent evaluation: add patient counts, principal paired effects with uncertainty, and a conclusion supported by the results.]

Keywords: Deformable image registration · PSMA PET/CT · Longitudinal imaging · Biomarker preservation.

1 Introduction

Prostate-specific membrane antigen (PSMA) PET/CT has become a central imaging modality for staging, treatment planning, radiopharmaceutical therapy, and post-therapy response assessment in prostate cancer [6]. In these settings, patients often undergo multiple scans over time, comprising a baseline scan before therapy and one or more follow-up scans. Accurate baseline-to-follow-up registration can support the comparison of disease burden, the localization of uptake changes, the assessment of lesion progression or response, dose accumulation, and quantitative longitudinal analysis [11]. For such analyses, the deformation should align anatomy without introducing artificial changes in PET-derived tumor burden measures. The Learn2Reg 2026 PSMAReg challenge [8, 7, 9] formalizes this problem as the estimation of a dense displacement field that warps each follow-up (moving) scan to its baseline (fixed) scan. The challenge evaluates registration accuracy using organ overlap and boundary distance, deformation regularity, and preservation of two PET-derived biomarkers: metabolic tumor volume (MTV) and total lesion glycolysis (TLG).

Deformable registration can be formulated as iterative optimization of a regularized objective functional [20, 1, 16], which can achieve high accuracy but is computationally demanding for whole-body images. Learning-based methods instead amortize this optimization into network weights and replace per-pair iteration with a single forward pass at inference [3, 5, 4]. We therefore build on LapIRN [15], whose coarse-to-fine Laplacian pyramid is well suited to recovering

the large deformations expected between longitudinal whole-body scans while maintaining smooth and invertible transformations.

This journal study builds on our earlier registration pipeline and asks how PET preservation and bone rigidity affect alignment, lesion quantification, and deformation quality. Preservation refers to changes introduced by warping the same scan; it does not require uptake or tumor burden to remain equal between longitudinal examinations. [TODO: Add the citation to the earlier paper and a concise statement of the journal extension.] The study is organized around three questions:

- What are the separate and combined effects of PET preservation and bone rigidity, and what does each PET loss term contribute?
- How do existing rigidity penalties compare in low-resolution whole-body CT, including their effects on bone shape and surrounding tissue?
- How does the strength of PET preservation change the balance between anatomical alignment, lesion preservation, and deformation quality?

The rigidity study evaluates existing approaches in this setting rather than introducing a new rigidity method.

2 Methods

2.1 Data and Preprocessing

We used the PSMAReg dataset of the Learn2Reg 2026 challenge [11], a longitudinal whole-body PSMA PET/CT cohort in which each subject provides a baseline scan and one or more follow-up scans. Each scan contains CT and PSMA PET channels; follow-up scans were treated as moving images and baseline scans as fixed images. The images are provided at $2.7344 \times 2.7344 \times 3.27$ mm resolution and cropped to $192 \times 192 \times 288$ voxels. We therefore applied no further geometric preprocessing. CT was clipped to $[-1000, 1500]$ HU and PET SUV to $[0, 20]$, followed by rescaling to $[0, 1]$.

The journal evaluation will use a patient-level train/validation/test partition of the labelled challenge training dataset, keeping every scan and registration pair from a patient in one partition. Registration models and the lesion nnU-Net will be trained only on training patients; validation patients will be used for all tuning and model selection. Previously unused patients will preferentially be reserved for testing, and prior patient exposure in pipeline development will be disclosed. [TODO: Freeze the split manifest; report patients, scans, pairs, and lesions per partition, eligibility criteria, pair direction, and prior exposure. Confirm whether a previously unused test subset is feasible.]

At test time, TotalSegmentator predictions and predictions from the training-only nnU-Net will supply the labels required by instance optimization (IO). Provided labels will be reserved for evaluation on test patients and will not enter IO or test-time iterate selection. [TODO: Document the provenance of the provided organ and lesion labels, the segmentation models and their training

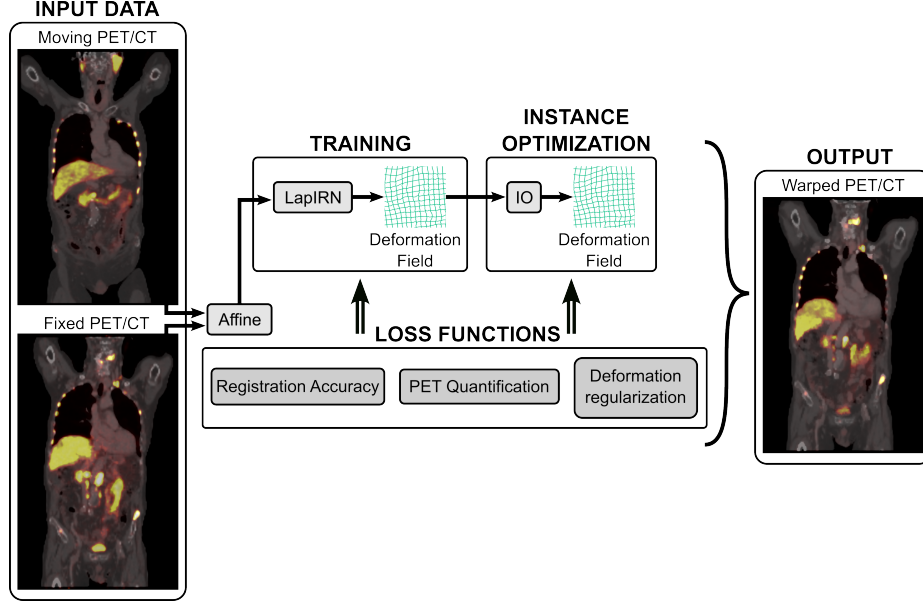


Fig. 1. Overview of the proposed registration pipeline. The moving (follow-up) and fixed (baseline) PET/CT scans are first coarsely aligned by an affine pre-registration, and the resulting pair is then passed to the coarse-to-fine LapIRN network, which predicts a dense deformation field (green grid). Finally, after the LapIRN stage, we refine this field per pair by instance optimization (IO) and resample the moving scan with it into the fixed space (warped PET/CT). Both stages are driven by the same objective, a weighted sum of three groups of terms: registration accuracy, PET quantification, and deformation regularization (Sec. 2.3).

data, and any dependence between evaluation annotations and predicted labels.]

Following the ConvexAdam [17] baseline provided by the challenge organizers, we additionally removed the patient bed and normalized the CT images.

2.2 Registration Pipeline

Our pipeline consists of CT-based affine pre-registration, LapIRN-based deformable registration, and optional test-time instance optimization (Fig. 1). Throughout, I denotes the moving PET image, M its lesion mask, φ_{aff} the affine pre-registration, φ_{θ} the network-predicted deformation, and φ their composition. W is the lesion mask warped into the fixed frame.

Step 1: Affine pre-registration. For each pair, ANTs [2] estimates an affine transform from the CT images alone, using body masks, CT windowing to $[-1000, 1000]$ HU, half-resolution images, and Mattes mutual information. The transform is converted to a dense displacement field and applied to the moving scan.

Step 2: Deformable registration. The deformable stage uses LapIRN [15], with three pyramid levels operating at quarter, half, and full resolution. Each level predicts a stationary velocity field, adds it to the upsampled velocity from the coarser level, and exponentiates the result by scaling and squaring to obtain φ_θ . To handle PSMA PET/CT, each sub-network receives four channels: fixed CT and PET, and affinely aligned moving CT and PET. The similarity loss is computed only on CT, because PET uptake may change after therapy; PET is used as input and in the preservation terms of Sec. 2.3. The levels are trained sequentially, with each finer level initialized from the previous one.

Step 3: Instance optimization. At test time, each pair is refined by optimizing a residual stationary velocity field on top of the network prediction. The residual is initialized to zero and exponentiated by scaling and squaring. The velocity formulation targets diffeomorphic transformations; discrete folding will nevertheless be measured explicitly. IO will use the objective appropriate to each experimental condition (Sec. 2.4). Its stopping and iterate-selection rule will be chosen on validation patients and frozen before testing; test evaluation labels will not be used to score or select iterates. [TODO: Specify optimized variables, learning rate, iteration budget, and stopping/selection rule, including which predicted-label quantities it may use.]

2.3 Registration Objective

We use a weighted sum of three loss groups, one per evaluation criterion:

$$\begin{aligned} \mathcal{L} = & \underbrace{\lambda_{\text{NCC}} \mathcal{L}_{\text{NCC}} + \lambda_{\text{DSC}} \mathcal{L}_{\text{DSC}}}_{\text{registration accuracy}} \\ & + \underbrace{\lambda_{\text{MTV}} \mathcal{L}_{\text{MTV}} + \lambda_{\text{MTV-J}} \mathcal{L}_{\text{MTV-J}} + \lambda_{\text{jac}} \mathcal{L}_{\text{jac}} + \lambda_{\text{TLG}} \mathcal{L}_{\text{TLG}}}_{\text{PET quantification}} \\ & + \underbrace{\lambda_{\text{rigid}} \mathcal{L}_{\text{rigid}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}} + \lambda_{\text{NDV}} \mathcal{L}_{\text{NDV}}}_{\text{deformation regularization}}. \end{aligned} \quad (1)$$

Each group is described below. All terms are evaluated on the composed transformation φ . Only the full-resolution level is trained with the complete objective. The two coarser levels (quarter and half resolution) use only the similarity, label, and regularization terms; thus $\lambda_{\text{MTV}} = \lambda_{\text{MTV-J}} = \lambda_{\text{jac}} = \lambda_{\text{TLG}} = \lambda_{\text{rigid}} = 0$. We omit the PET and bone-specific terms at these resolutions because downsampled lesions and thin ribs span only a few voxels, so volume ratios and per-structure rigid fits mainly reflect interpolation rather than anatomy.

Registration Accuracy. Image similarity is measured using local normalized cross-correlation (NCC), computed as a multi-resolution sum over as many scales as the respective pyramid level has resolutions, i.e. one scale at the coarsest and three at the full-resolution level. NCC is computed on the CT channel only,

since PET uptake can change substantially between time points, and a PET similarity term could therefore encourage matching disease-related uptake rather than anatomy. We additionally use a DSC loss for weak supervision from the CT organ labels.

Preservation of PET quantification. PET preservation compares the moving scan before and after transformation, using the same source lesion definition. For the differentiable training and IO objective, let $\tilde{I} = W_{\text{lin}}[I]$ and $\tilde{M} = W_{\text{lin}}[M]$ denote the linearly warped PET image and soft lesion mask. We compute

$$\mathcal{L}_{\text{MTV}} = \left(\frac{\left| \sum_x \tilde{M}(x) - \sum_x M(x) \right|}{\sum_x M(x) + 10^{-5}} \right)^2. \quad (2)$$

$$\mathcal{L}_{\text{TLG}} = \frac{\left| \sum_x \tilde{I}(x) \tilde{M}(x) - \sum_x I(x) M(x) \right|}{\sum_x I(x) M(x) + 10^{-5}}. \quad (3)$$

These losses enter Eq. (1) and deliberately differ from the official evaluation metrics of Sec. 2.5. Nearest-neighbour mask interpolation is piecewise constant with respect to the displacement and therefore provides no useful gradient for registration; linear interpolation instead supplies a differentiable soft boundary. The additive denominator stabilizes optimization for empty or very small masks, and squaring the relative MTV error gives a smooth quadratic penalty that emphasizes larger volume errors. Training uses clipped, normalized PET to keep gradient scales bounded, whereas official TLG evaluation uses PET in SUV units. [TODO: Write the exact equations for the two Jacobian-based PET terms and verify transformation direction, Jacobian convention, reduction, and handling of empty or tiny lesions against the final implementation.] All terms are evaluated on the composed transformation φ and the original moving lesion mask M .

To additionally constrain the deformation within the lesion, we use $\mathcal{L}_{\text{MTV-J}}$, which penalizes deviations of the mean Jacobian determinant over the warped lesion region W from one. A third term, $\mathcal{L}_{\text{jac}}$, penalizes per-voxel deviations of the Jacobian determinant from one and therefore discourages local compensating compression and expansion.

The full-objective condition uses all four PET preservation terms during training and instance optimization; the ablations remove the specified terms.

Bone rigidity. Many lesions lie within skeletal structures, where rigid motion is physiologically plausible and preserves both volume and intensity ($\det J_\varphi = 1$). We therefore penalize departures from rigid motion within each skeletal structure. For the 61 skeletal TotalSegmentator labels, we fit the displacement of each structure independently with a rigid transformation using closed-form Kabsch alignment [10, 19] and penalize the residual. This constrains the deformation within each structure while allowing independent motion between structures.

Deformation regularization. We regularize the displacement field using a diffusion term, $\mathcal{L}_{\text{smooth}}$, defined as the mean squared spatial gradient of the displacement, together with a penalty for non-diffeomorphic voxels. For the latter, we differentially reproduce the challenge’s non-diffeomorphic volume (NDV) metric, which decomposes each voxel into six tetrahedra and accumulates the negative part of their Jacobian determinants over the body mask [12]. The resulting $\mathcal{L}_{\text{NDV}}$ directly corresponds to the scored percentage of non-diffeomorphic volume and provides gradients at the locations where folding occurs.

2.4 Experimental Design and Model Selection

All experiments below are planned. Settings will be selected on validation patients and frozen before evaluation on test patients. The split, endpoints, and analysis rules will be fixed before running the core experiments. [TODO: Define the validation selection criterion, sweep ranges, compute budgets, and repeat/seed policy. Do not reuse the previous surrogate-set ranking as journal test evidence.]

PET and rigidity ablations.

- Train the 2×2 conditions: neither PET preservation nor rigidity, PET only, rigidity only, and both. Retain the same registration-accuracy and general regularization terms.
- Within the full objective, remove each of $\mathcal{L}_{\text{MTV}}$, $\mathcal{L}_{\text{MTV-J}}$, $\mathcal{L}_{\text{jac}}$, and $\mathcal{L}_{\text{TLG}}$ individually.
- Prioritize matched training for the central claims. Specify each condition’s training and IO objective separately; identify any IO-only ablation as such.
- Report paired effects of PET, rigidity, and their combination on alignment, lesion preservation, and deformation quality; include the PET–rigidity interaction if estimable under the chosen analysis.

Comparison of rigidity penalties. Compare no rigidity, Staring with all three components [18], Staring with mask erosion, and the existing Kabsch rigid-fit penalty [10]. Tune or sweep each method’s weight separately; equal numerical weights need not imply comparable constraint strength.

- Document the three Staring components, their scaling, derivative stencils, boundary handling, and coordinate units. Specify the erosion rule and report bones/voxels retained.
- Use alignment versus physical within-bone distance distortion as the main comparison, with folding alongside. Report bone rigidity error (BRE) only as a secondary endpoint because it closely matches the Kabsch objective.
- Measure direct stencil leakage separately from the effect on surrounding tissue. Kabsch avoids direct stencil leakage, but shared deformation parameters and regularization can still influence soft tissue.
- Scope conclusions to the tested resolutions, implementations, and weight ranges.

Accuracy–preservation trade-off. Run a small PET-weight sweep including zero and the current setting. [TODO: Specify whether one multiplier scales the whole PET group, which relative term weights remain fixed, and whether each setting is retrained or changes IO only.] Plot alignment against lesion preservation and report deformation quality for every setting. Select operating points on validation patients and evaluate frozen settings on the test partition, without choosing a preferred setting from test results.

Baselines. Re-evaluate the initial alignment, affine registration, NiftyReg [14, 13], and ConvexAdam [17] on the new test partition under the same evaluation definitions. Report the full method with and without IO. [TODO: Freeze baseline configurations, validation tuning allowances, and runtime measurement. The previous container and surrogate-validation configurations are not journal comparison groups.]

2.5 Evaluation Endpoints and Statistical Analysis

Official PET preservation metrics. For the official evaluation, PET is expressed in SUV and resampled using linear interpolation, while the binary lesion mask is resampled using nearest-neighbour interpolation. MTV is the physical volume of all lesion voxels. TLG is the sum of their PET intensities multiplied by the physical voxel volume. Both biomarkers are measured before and after registration.

Following the official challenge implementation, the error for either biomarker Q is its absolute percentage change:

$$E_Q(\%) = 100 \frac{|Q_{\text{after}} - Q_{\text{before}}|}{\max(Q_{\text{before}}, 10^{-6})}, \quad Q \in \{\text{MTV}, \text{TLG}\}. \quad (4)$$

Both official errors are aggregate scan-level measures: biomarker increases in some lesions and decreases in others can cancel before the absolute difference is taken, yielding a small aggregate error despite substantial individual lesion errors. Because PET and the lesion mask are interpolated separately before multiplication, E_{TLG} captures both deformation-related changes and resampling effects at the lesion boundary.

Metric implementation validation. Before model comparison, deterministic controls verify the complete metric implementation on a real PET/CT case. An identity field must preserve MTV, TLG, Dice, HD95, and within-bone distances and produce no folding. A whole-voxel translation, chosen to avoid field-of-view clipping, must preserve MTV, TLG, and within-bone distances, produce no folding, and recover the prescribed displacement in physical units. A known uniform stretch must produce the expected relative within-bone distance change. To characterize resampling alone, we additionally apply translations of 0.25, 0.49, 0.50, 0.51, and 0.75 voxels while holding the underlying lesion and transformation

geometry fixed. For each shift, the official calculation above is compared with a control that first forms IM , linearly warps this product once, and sums the result over the whole image. The 0.49- and 0.51-voxel conditions bracket the nearest-neighbour tie at exactly half a voxel. These controls define a numerical resampling baseline and are not treated as registration-performance experiments.

- **Alignment:** per-bone and soft-tissue organ Dice and surface distance; near-bone landmarks only if independently available. Define aggregation over structures and handling of absent or truncated anatomy.
- **PET preservation:** retain the official aggregate MTV/TLG errors as primary endpoints. Additionally compute absolute MTV/TLG error for each source lesion, summarize lesion errors within each patient before cohort-level aggregation, and report a predeclared upper-tail statistic stratified by bone association and lesion size. Compare aggregate and per-lesion summaries to assess cancellation without changing the official endpoint.
- **Lesion measurement:** define lesions on the source scan and track them through the transformation. Retain one analysis row per registration pair and lesion, but give patients equal weight in cohort summaries so patients with many scans or lesions do not dominate. Predefine size bins, bone-association rules, minimum denominators, empty-mask handling, field-of-view exclusions, SUV units, and interpolation. Distinguish resampling effects from deformation effects using a specified control.
- **Bone shape:** measure distortion of sampled within-bone point-pair distances in physical coordinates, covering short and long distances. Predefine sampling, distance bins, and normalization; report BRE secondarily.
- **Deformation quality:** report NDV/folding inside bones, in an outside boundary band, and across the body. Specify masks, band width in physical units, denominators, and consistent units.
- **Boundary mechanism:** quantify the fraction of stencil samples outside bone and the fraction of bones/voxels retained after erosion. State how bones that disappear after erosion are handled.
- **Uncertainty:** report paired method effects with patient-level confidence intervals, accounting for multiple pairs and lesions per patient. A candidate is patient-cluster bootstrap resampling that retains all within-patient observations and method pairing.

[TODO: Freeze primary endpoints and contrasts, the upper-tail statistic, patient weighting, uncertainty procedure, and multiplicity handling before test evaluation. Report evaluable patient/pair/lesion counts and exclusions.]

2.6 Optional Supporting Analyses

- **Gradient analysis:** one loss-pair cosine matrix, selected IO trajectories, and representative maps. Distinguish displacement-space maps from gradients in optimized velocity coordinates; mask near-zero gradients and report weighted magnitudes. Local cosine similarity is not evidence of a final performance trade-off.

- **External cohort:** include only if feasible. Combine automatic measurements with blinded, randomized clinician review; add sparse landmarks if available. Separate IO labels from independent evaluation. Distinguish re-sampling from deformation effects.

These analyses are optional; independent evaluation and the core experiments take priority.

2.7 Implementation Details

Augmentations were chosen to mimic longitudinal variability: left–right flipping ($p = 0.5$), CT intensity shifting by ± 0.02 in the normalized window (about ± 50 HU), CT scaling from 0.9 to 1.1, and PET scaling from 0.85 to 1.15 without additive shifts. To simulate a field-of-view mismatch, we crop up to 40 axial slices from either end of both scans, followed by independent moving and fixed crops of up to 10 additional slices.

Training configuration to confirm. The previous pipeline used sequential Adam training for 60k, 60k, and 120k steps, learning rates of $3 \cdot 10^{-4}$, $2 \cdot 10^{-4}$, and $2.5 \cdot 10^{-4}$, batch size one, and gradient accumulation over four steps. These are starting settings for the journal retraining, not completed journal experiments.

- Confirm the warmup and coarse-level freezing schedule, augmentations, loss weights, and random seeds. Keep the training budget and shared settings matched across central ablations.
- Record mixed precision, transformation precision, final software versions, hardware, training time, and per-pair inference/IO time for the new runs.
- Confirm which pyramid levels use PET and rigidity terms in each condition.

3 Results

Journal draft: model-comparison experiments are pending. Except for the deterministic metric-validation controls reported below, all tables and figures are planned outputs rather than findings. The previous evaluation and leaderboard results have been removed because they used a different protocol.

3.1 Metric Implementation Validation

On PSMAReg subject 0006 at time point 02, the identity control produced zero MTV and TLG error, zero within-bone distance distortion, zero folding, Dice of one, and HD95 of zero. A non-clipping whole-voxel translation likewise preserved MTV, TLG, and within-bone distances exactly, produced no folding, and recovered the prescribed 26.91 mm displacement. For subvoxel translations of 0.25, 0.49, 0.50, 0.51, and 0.75 voxels, the official absolute TLG errors were 1.50%, 2.94%, 2.95%, 2.76%, and 1.41%, respectively, despite the transformation being

a pure translation. In contrast, linearly warping the pre-multiplied PET-mask product changed its global sum by at most $1.4 \times 10^{-4}\%$. MTV was preserved on either side of the nearest-neighbour tie (0.49 and 0.51 voxels); at exactly 0.50 voxels, tie-dependent nearest-neighbour assignment changed MTV by 0.16%. These controls confirm the implementation and quantify a case-specific interpolation floor; they do not measure registration accuracy or biological change.

3.2 Independent-Test Comparison

- Report the final cohort, exclusions, and evaluable structures/lesions.
- Compare baselines and the full model with and without IO on the frozen test partition (Table 1).
- Add paired patient-level effects and uncertainty; describe alignment, preservation, folding, and runtime without selecting a winner from test results.

Table 1. Planned independent-test comparison. All entries are pending; “pending” is not a numerical result. Report patient-level uncertainty and paired effects in the final table/text. Per-lesion and regional endpoints are detailed separately.

Method	Dice	HD95	MTV/TLG	NDV	Runtime
Initial	pending	pending	pending	pending	pending
Affine	pending	pending	pending	pending	pending
NiftyReg	pending	pending	pending	pending	pending
ConvexAdam	pending	pending	pending	pending	pending
Full model, no IO	pending	pending	pending	pending	pending
Full model + IO	pending	pending	pending	pending	pending

3.3 Separate and Combined Effects of PET Preservation and Rigidity

- Fill the matched-training 2×2 comparison and the four PET-term removal conditions (Table 2).
- Report the official aggregate errors alongside patient-level summaries of per-lesion absolute errors; quantify whether aggregate cancellation hides large individual errors.
- Show absolute and upper-tail errors by lesion size and bone association (Fig. 2).

3.4 Rigidity Comparison and Boundary Effects

- Compare each rigidity method over its own weight range: bone/organ alignment, distance distortion, regional folding, and secondary BRE.
- Relate stencil leakage and erosion retention to the observed outcomes without assuming a mechanism in advance (Fig. 3).
- Include representative bone-boundary examples under a documented case-selection rule.

Table 2. Planned matched-training ablations. Accuracy and general regularization terms are retained. The endpoint groups are alignment, aggregate/per-lesion PET preservation, and deformation quality. Results and patient-level paired effects are pending; any additional IO-only ablations must be reported separately.

Condition	PET terms	Rigidity	Results
Neither	None	Off	Pending
PET only	All four	Off	Pending
Rigidity only	None	On	Pending
Both (full)	All four	On	Pending
Full minus $\mathcal{L}_{\text{MTV}}$	Other three	On	Pending
Full minus $\mathcal{L}_{\text{MTV-J}}$	Other three	On	Pending
Full minus $\mathcal{L}_{\text{jac}}$	Other three	On	Pending
Full minus $\mathcal{L}_{\text{TLG}}$	Other three	On	Pending

Planned panels: aggregate versus per-lesion MTV/TLG error; absolute error distributions; upper-tail summaries by lesion size and bone association.

Fig. 2. Planned lesion-preservation analysis for the matched PET and rigidity ablations. Compare each moving scan before and after warping, with patient-level uncertainty. State subgroup counts and the predeclared upper-tail statistic. No experimental data are shown.

Planned panels: bone/organ alignment versus physical distance distortion over method-specific weights; folding inside bone and in the outer band; stencil leakage and erosion retention; representative boundary maps.

Fig. 3. Planned comparison of no rigidity, Staring, Staring with erosion, and Kab-sch. Report regional folding alongside alignment and distance distortion, and show boundary-mechanism measurements separately. BRE is secondary. No experimental data are shown.

3.5 Accuracy–Preservation Trade-off

- Show the predeclared PET-weight settings and their patient-level uncertainty (Fig. 4).
- Identify the validation-selected operating point and report its frozen test performance.
- Discuss whether preservation changes alignment or folding, including heterogeneous lesion responses.

Planned panels: alignment versus per-lesion MTV and TLG preservation across a small PET-weight sweep, including zero and the current setting; corresponding deformation quality.

Fig. 4. Planned accuracy–preservation trade-off. Mark the operating point selected on validation patients and distinguish validation curves from frozen-setting test results. Include patient-level uncertainty and identify whether settings change matched training or IO only. No experimental data are shown.

3.6 Optional Supporting Results

[TODO: Include gradient analysis and/or external-cohort results only if completed; otherwise remove this subsection. Do not imply external validation from surrogate measurements alone.]

4 Discussion

Discussion outline; replace with evidence-based interpretation after the core experiments.

- **Main answer:** summarize the separate and combined effects of PET preservation and bone rigidity, with the principal effect sizes and uncertainty.
- **Lesion quantification:** explain whether aggregate preservation conceals per-lesion errors, which lesion groups benefit, and whether any groups show worse upper-tail errors. Preserving a warped scan’s quantification does not establish clinical response accuracy or imply unchanged disease across time.
- **Rigidity:** interpret alignment versus distance distortion alongside folding and soft-tissue effects. Discuss stencil leakage and erosion retention only to the extent supported by the measurements. Avoid claiming a new rigidity method or general superiority beyond the tested setting.

- **Trade-off:** explain the practical consequences of the PET-weight sweep and how validation-only selection determines the reported operating point.
- **Evaluation validity:** discuss prior patient exposure, label provenance, possible dependence between predicted and evaluation labels, and uncertainty with repeated pairs/lesions. State any limitations of the independent test partition that remain after retraining.
- **Practical limitations:** low resolution, small lesions and thin bones, interpolation and field-of-view effects, segmentation dependence, and compute/runtime. Separately interpolating PET and a nearest-neighbour lesion mask introduces a non-zero, subvoxel-phase-dependent TLG error even for rigid translations. Because the same estimator enters the optimization objective, the model can reduce the measured interpolation error through its deformation; such a reduction must not automatically be interpreted as improved physical biomarker preservation. Interpret effects relative to the re-sampling control, and do not assume theoretical diffeomorphism eliminates discrete folding.
- **Scope and generalization:** distinguish the core labelled-cohort evidence from optional gradient analyses and external-cohort evidence, if available.

Conclusion to write after evaluation. [TODO: State what the independent evaluation supports about alignment, lesion preservation, and bone rigidity, including the main trade-off and remaining limitation.]

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References

1. Avants, B., Epstein, C., Grossman, M., et al.: Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain. *Medical Image Analysis* **12**(1), 26–41 (2008), special Issue on The Third International Workshop on Biomedical Image Registration - WBIR 2006
2. Avants, B.B., Tustison, N.J., Song, G., et al.: A reproducible evaluation of ants similarity metric performance in brain image registration. *NeuroImage* **54**(3), 2033–2044 (2011)
3. Balakrishnan, G., Zhao, A., Sabuncu, M.R., et al.: An unsupervised learning model for deformable medical image registration. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (June 2018)
4. Chen, J., Frey, E.C., He, Y., et al.: Transmorph: Transformer for unsupervised medical image registration. *Medical Image Analysis* **82**, 102615 (2022)
5. Dalca, A.V., Balakrishnan, G., Guttag, J., et al.: Unsupervised learning for fast probabilistic diffeomorphic registration. In: Frangi, A.F., Schnabel, J.A., Davatzikos, C., et al. (eds.) *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*. pp. 729–738. Springer International Publishing, Cham (2018)

6. Fendler, W.P., Eiber, M., Beheshti, M., et al.: 68Ga-PSMA PET/CT: Joint EANM and SNMMI procedure guideline for prostate cancer imaging: version 1.0. *European Journal of Nuclear Medicine and Molecular Imaging* **44**(6), 1014–1024 (Jun 2017)
7. Hansen, L., Heyer, W., Großbröhmer, C., et al.: Learn2reg 2024: New benchmark datasets driving progress on new challenges. *Machine Learning for Biomedical Imaging* **3**, 775–791 (2025)
8. Hering, A., Hansen, L., Mok, T.C.W., et al.: Learn2reg: Comprehensive multi-task medical image registration challenge, dataset and evaluation in the era of deep learning. *IEEE Transactions on Medical Imaging* **42**(3), 697–712 (2023)
9. Jeblick, K., Schönberg, S.O., et al.: A whole-body PSMA-PET/CT dataset with manually annotated tumor lesions (PSMA-PET-CT-Lesions). *The Cancer Imaging Archive* (2024), version 1
10. Kabsch, W.: A solution for the best rotation to relate two sets of vectors. *Foundations of Crystallography* **32**(5), 922–923 (1976)
11. Learn2Reg Consortium: Learn2Reg 2026 – PSMAReg: Registration of pre-therapy and follow-up whole-body PSMA PET/CT scans. Codabench challenge (2026), <https://www.codabench.org/competitions/15724/>
12. Liu, Y., Chen, J., Wei, S., et al.: On finite difference jacobian computation in deformable image registration. *International Journal of Computer Vision* **132**(9), 3678–3688 (Sep 2024)
13. Modat, M., Cash, D.M., Daga, P., et al.: Global image registration using a symmetric block-matching approach. *Journal of Medical Imaging* **1**(2), 024003 (2014)
14. Modat, M., Ridgway, G.R., Taylor, Z.A., et al.: Fast free-form deformation using graphics processing units. *Computer Methods and Programs in Biomedicine* **98**(3), 278–284 (2010), hP-MICCAI 2008
15. Mok, T.C.W., Chung, A.C.S.: Large deformation diffeomorphic image registration with laplacian pyramid networks. In: Martel, A.L., Abolmaesumi, P., Stoyanov, D., et al. (eds.) *Medical Image Computing and Computer Assisted Intervention – MICCAI 2020*, pp. 211–221. Springer International Publishing, Cham (2020)
16. Rueckert, D., Sonoda, L., Hayes, C., et al.: Nonrigid registration using free-form deformations: application to breast mr images. *IEEE Transactions on Medical Imaging* **18**(8), 712–721 (1999)
17. Siebert, H., Hansen, L., Heinrich, M.P.: Fast 3d registration with accurate optimisation and little learning for learn2reg 2021. In: Aubreville, M., Zimmerer, D., Heinrich, M. (eds.) *Biomedical Image Registration, Domain Generalisation and Out-of-Distribution Analysis*, pp. 174–179. Springer International Publishing, Cham (2022)
18. Staring, M., Klein, S., Pluim, J.P.W.: A rigidity penalty term for nonrigid registration. *Medical Physics* **34**(11), 4098–4108 (2007)
19. Umeyama, S.: Least-squares estimation of transformation parameters between two point patterns. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **13**(4), 376–380 (1991)
20. Vercauteren, T., Pennec, X., Perchant, A., et al.: Diffeomorphic demons: Efficient non-parametric image registration. *NeuroImage* **45**(1, Supplement 1), S61–S72 (2009)